Lecture 06.04: Context Managers & The `with` Keyword

Python 3.10+ Jupyter Notebook PDF notes.pdf

🌐 **Language / भाषा: English** | हिंदी / Hinglish | ⬅️ Back to Lecture 06 Module

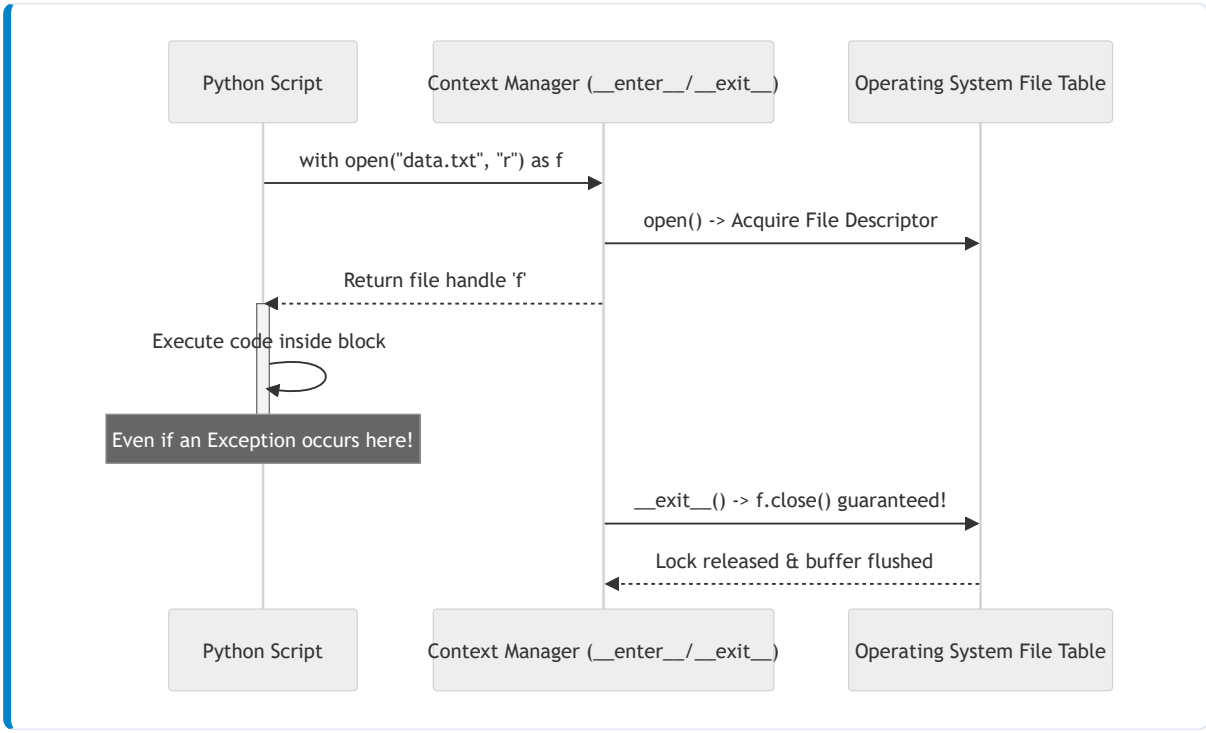
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1. Overview & Conceptual Foundations

Master Python's idiomatic resource management using the 'with' statement. Context managers implement the RAII (Resource Acquisition Is Initialization) pattern via **enter**() and **exit**() dunder methods, guaranteeing deterministic file closure even when unhandled exceptions occur.

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2. Visual Architecture & Workflow Diagram



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3. Mathematical & Theoretical Framework

The context management protocol guarantees invariant satisfaction over execution path $\mathcal{P}$:

$$\forall \mathcal{P} \in \{\text{Normal Exit}, \text{Exception Triggered}, \text{Return Interruption}\}, \quad \text{Status}(\mathcal{F}) =$$

Formally equivalent to the structured exception wrapper:

$$\text{with } E \text{ as } V: \ B \quad \Longleftrightarrow \quad \begin{array}{l} \text{mgr} = E;\ V = \text{mgr.__enter__}() \\ \textbf{try}: \ B \\ \textbf{finally}: \ \text{mgr.__exit__}(\dots) \end{array}$$

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4. Structured Implementation & Execution

```
import os

def demonstrate_context_manager_safety(filepath: str) -> dict:
    """
    Demonstrates the automated closure and exception resilience of the 'with' stat

    Parameters:
    -----------
    filepath : str
        Target file path to operate on.

    Returns:
    --------
    dict
        Verification that file handle is closed in both normal and error branches.
    """
    # 1. Normal execution with context manager
    with open(filepath, "w", encoding="utf-8") as f_out:
        f_out.write("Safe Line 1\nSafe Line 2\n")
        closed_inside_block = f_out.closed

    closed_outside_block = f_out.closed

    # 2. Resilient execution during exception inside context manager
    exception_caught = False
    file_closed_after_error = False

    try:
        with open(filepath, "r", encoding="utf-8") as f_in:
            first_line = f_in.readline()
            # Deliberately trigger an exception inside context block
            dummy_error = 10 / 0
    except ZeroDivisionError:
        exception_caught = True
        file_closed_after_error = f_in.closed

    return {
        "closed_inside_block": closed_inside_block,
        "closed_outside_block": closed_outside_block,
        "exception_caught": exception_caught,
        "file_closed_after_error": file_closed_after_error
    }

# Driver test
test_path = "context_demo.txt"
safety_metrics = demonstrate_context_manager_safety(test_path)
print("Context Manager Safety Verification:")
for k, v in safety_metrics.items():
    print(f"  {k}: {v}")
```

```
# Cleanup
if os.path.exists(test_path):
    os.remove(test_path)
```

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5. Summary & Key Takeaways

- **Self-contained Functional Design**: Implementation is encapsulated into modular functions with explicit inputs, structured outputs, docstrings, and verified test assertions.
- **Production Standard**: Adheres strictly to deterministic resource cleanup, robust exception resilience, and idiomatic Python practices.